

Replication Plan

NukeLM: Pre-Trained and Fine-Tuned Language Models for the Nuclear and Energy Domains

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Auto-generated Replication Blueprint

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Paper Overview

NukeLM presents domain-specific language models for the nuclear science and energy domains. Starting from general-purpose pre-trained models (RoBERTa base/large, SciBERT), the authors perform continued pre-training (domain-adaptive pre-training, DAPT) on a large corpus of nuclear/energy-domain text (OSTI abstracts, NRC documents, DOE reports). The models are then fine-tuned on downstream NLP tasks: named entity recognition (NER), relation extraction (RE), and text classification in the nuclear domain. The paper demonstrates that domain-specific pre-training significantly improves performance on these tasks.

Detailed Workflow

Phase 1: Corpus Collection and Pre-processing

1.1 Collect pre-training corpus.

- OSTI.gov abstracts: query the OSTI API for nuclear science and energy technology abstracts.
- NRC ADAMS documents (publicly available regulatory documents).
- DOE technical reports from SciTech Connect.
- Total corpus size: $\sim$ millions of documents / several GB of text.

1.2 Pre-process text.

- Clean HTML/XML artifacts, normalize whitespace, remove duplicates.
- Sentence tokenization and subword tokenization (matching RoBERTa/SciBERT tokenizer).

Phase 2: Domain-Adaptive Pre-Training

2.1 Load base models.

- RoBERTa-base (125M params), RoBERTa-large (355M params), SciBERT.
- Use HuggingFace Transformers `AutoModelForMaskedLM`.

2.2 Continued pre-training with MLM objective.

- Masked language modeling on the nuclear domain corpus.
- Training parameters: batch size, learning rate, warm-up steps, total steps (as specified in paper).
- Use distributed training (multi-GPU) for efficiency.

2.3 Save domain-adapted model checkpoints (NukeLM-base, NukeLM-large).

Phase 3: Downstream Task Fine-Tuning

3.1 Named Entity Recognition (NER).

- Prepare NER training data with nuclear domain entity tags (e.g., material, isotope, facility).
- Add a token classification head on top of NukeLM.
- Train and evaluate (precision, recall, F1 per entity type).

3.2 Relation Extraction (RE).

- Prepare annotated relation data.
- Fine-tune with entity marker strategy or similar.
- Evaluate F1 score.

3.3 Text Classification.

- Classify documents into nuclear sub-domains.
- Add a sequence classification head; fine-tune and evaluate accuracy/F1.

Phase 4: Comparison and Validation

4.1 Compare NukeLM vs. base RoBERTa, SciBERT, and BioBERT on all downstream tasks.

4.2 Reproduce the paper’s tables of F1 scores and accuracy.

4.3 Statistical significance tests (bootstrap or paired t-test).

Datasets

Dataset / Source	Type	Location / Notes
OSTI.gov abstracts	Pre-training corpus	https://www.osti.gov/api/v1/ (OSTI API)
NRC ADAMS documents	Pre-training corpus	https://www.nrc.gov/reading-rm/adams.html
DOE SciTech Connect	Pre-training corpus	https://www.osti.gov/scitech/
Nuclear NER dataset	Fine-tuning (la-beled)	Described in paper; may require manual annotation or author-provided data
Nuclear RE dataset	Fine-tuning (la-beled)	Described in paper; check supplementary materials
Nuclear text classification	Fine-tuning (la-beled)	Described in paper; OSTI category labels

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
HuggingFace Transformers	≥ 4.15	Model loading, training, fine-tuning https://huggingface.co/docs/transformers/
HuggingFace Datasets	latest	Dataset loading and preprocessing
PyTorch	≥ 1.10	Deep learning backend
RoBERTa (base/large)	pre-trained	Base models from HuggingFace Hub

Tool / Library	Version	Purpose / URL
SciBERT	pre-trained	Scientific domain base model https://huggingface.co/allenai/scibert_scivocab_uncased
seqeval	latest	NER evaluation metrics
scikit-learn	≥ 0.24	Classification metrics
Weights & Biases / Tensor-Board	latest	Training monitoring (optional)

Hardware Requirements

- **Pre-training:** Multiple NVIDIA A100 or V100 GPUs (32 GB+); estimated 1–7 days.
- **Fine-tuning:** Single GPU with ≥ 16 GB VRAM.
- **Storage:** ≥ 100 GB for corpus and model checkpoints.